

MuSR Blackboard Truthful-Report Study

q=3, DeepSeek-V4-Flash

Preliminary Analysis Report

Control Efficiency of Multi-Agent Systems

7 September 2026

Important status. The supplied aggregation is incomplete: 720/750 planned episodes (96%) are present. The package validation is marked INVALID because 30 target episode keys are missing. There are no recorded failed or aborted episodes. Results below are therefore suitable for scientific diagnosis and experiment design, but should not yet be treated as the final production estimate.

1 Study design

This study tests a **truthful strategic-report controller** on the MuSR Team Allocation blackboard. The controller never fabricates evidence; on an active control round it injects canonical truthful reports selected by the target-preserving strategy. The study uses a binary control variable $U_k \in \{0, 1\}$, so the main feedback-control quantity remains

$$T_\pi = I(U_k; n_{k+1} \mid n_k).$$

Parameter	Value
Gameplay model	DeepSeek-V4-Flash via DeepInfra
Prompt	relational_blackboard_ballot@3
Task	MuSR Team Allocation task_001
Population	$N = 24$
Rounds	10
Blackboard sample size	$q = 3$
Sensor sample size	$q_c = 12$
Controller policy	soft target, $\beta = 4$, $\theta = 0.5$
Controller	truthful_strategic_report, dawn only
Persistence	$\rho \in \{0.70, 0.775, 0.85, 0.925, 1.0\}$
Budgets	$b \in \{3, 6, 9, 12, 15, 18, 21\}$
Arms	no control, truth control, false control
Repetitions	10 planned per cell
Board lifetime	1 round
Estimator bootstrap	1000 whole-episode resamples
Permutation nulls	1000

Table 1: Main study parameters recovered from the resolved configurations.

The planned grid contains 75 cells: 5 no-control persistence cells and $5 \times 7 = 35$ cells for each controlled arm. The supplied archive contains 48/50 no-control episodes, 336/350 truth-control episodes, and 336/350 false-control episodes. Thus missingness is numerically balanced between truth and false control (14 episodes missing in each arm), although several individual cells have only 8 or 9 repetitions.

2 Executive findings

1. **The truthful controller clearly changes behavior.** Truth-directed control raises the final truth share to about 0.76–0.84 across persistence values when averaged over b . False-directed truthful disclosure suppresses truth relative to no control and can make the false target the plurality outcome in a substantial fraction of episodes.
2. **False control is strongly non-monotone in budget.** Aggregated over persistence, the strongest false-target behavior occurs near $b = 9$ and $b = 18$, not at the largest budget. At $b = 9$, the false target averages 0.446 final share and is the final plurality in 56.3% of episodes; at $b = 18$ the corresponding values are 0.439 and 52.1%.
3. **Persistence no longer produces a simple “high $\rho = \text{uncontrollable}$ ” picture.** No-control truth improves strongly with ρ , reaching 0.863 at $\rho = 1$. Yet false control still lowers final truth to about 0.49 when averaged over budgets at $\rho = 1$. The new public truthful-report channel therefore remains behaviorally consequential even under perfect private-evidence persistence.
4. **Immediate susceptibility is much stronger for false control.** Whole-cell χ averages 0.0875 for false control versus 0.0272 for truth control. The bootstrap 95% interval excludes zero positively in 26/35 false cells but only 4/35 truth cells.
5. **Raw T_π is large but the permutation null is also large.** Raw target CMI averages 0.563 bits under false control and 0.447 bits under truth control, but the corresponding permutation-null means are approximately 0.527 and 0.470 bits. Only 2/35 false-control cells and 0/35 truth-control cells have permutation $p < 0.05$. The raw CMI surface should therefore be treated as descriptive until the finite-sample/state-conditioning bias is resolved.
6. **The communication channel shows strong diminishing returns.** On ACT rounds, controller-report exposures rise from 23.7 at $b = 3$ to 52.7 at $b = 21$, but target adoptions rise only from 2.56 to 3.51. Target adoption per exposure falls from roughly 10.8% to 6.7%.
7. **Higher b crowds out peer exposure while increasing evidence coverage.** Mean peer-report exposures fall from about 42.4 to 30.4 per round as b rises from 3 to 21, while active and historical evidence coverage increase. This is a direct control-efficiency trade-off: more controller information reaches the population, but it occupies attention that would otherwise go to peer communication.
8. **Participants almost never ask questions in this run.** REQUEST counts are only about 0.14–0.38 per round depending on persistence and about 0.2–0.3 per round across budgets. The current experiment is therefore primarily a report-exchange system, which strongly motivates the planned explicit cooperation/REQUEST on-off study.
9. **Thermodynamic efficiency η_{th} is still unsupported.** The affinity h cannot be calibrated from the current blackboard actuator records, so all η_{th} cells are undefined. J_c and sensing terms are present, but a new blackboard-compatible affinity calibration is still required.

3 Behavioral phase diagrams

3.1 No-control baseline

The autonomous dynamics become increasingly truth-favoring as persistence rises. Final truth shares are approximately 0.533, 0.700, 0.644, 0.667, and 0.863 for $\rho = 0.70, 0.775, 0.85, 0.925, 1.0$, respectively. The non-monotonic middle values are consistent with only 9–10 episodes per persistence point.

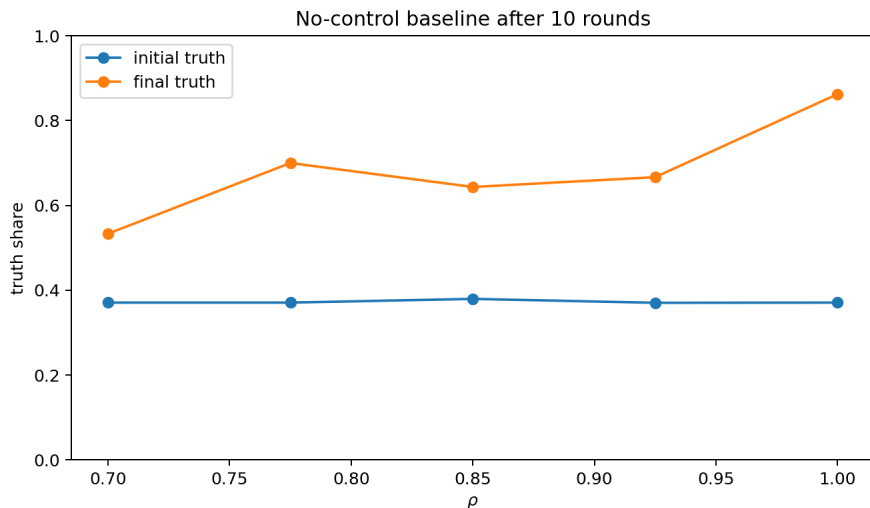


Figure 1: No-control truth share before and after the 10-round episode.

3.2 Truth control

Truth-directed truthful reports are effective over almost the entire grid. Final truth share is generally around 0.7–0.9. Averaged over budgets, final truth is 0.775 at $\rho = 0.70$, 0.760 at 0.775, 0.822 at 0.85, 0.818 at 0.925, and 0.839 at 1.0. At $\rho = 1$ the no-control baseline is already 0.863, so truth control offers no average improvement and can slightly underperform through attention crowding.

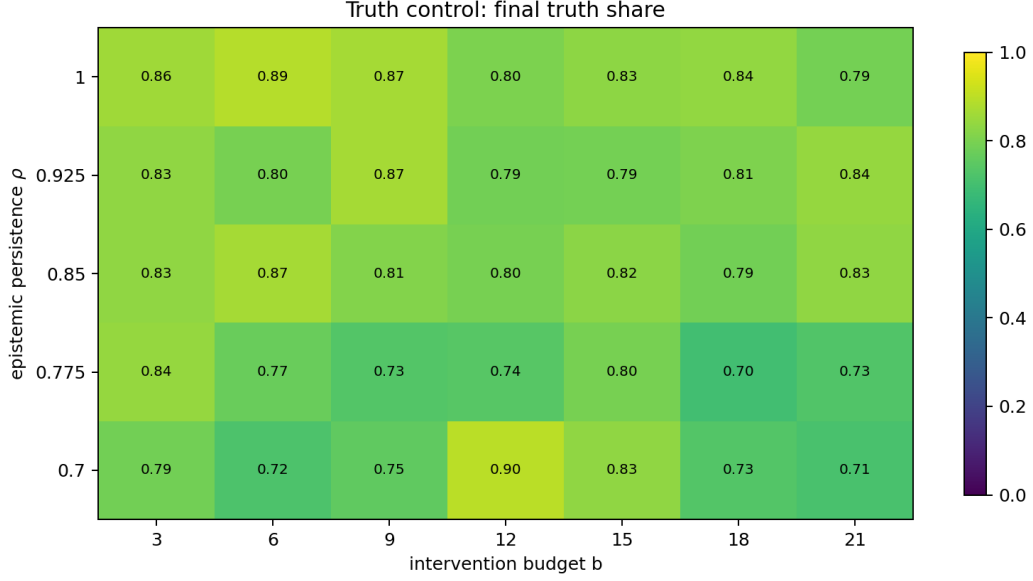


Figure 2: Final truth share under truth control.

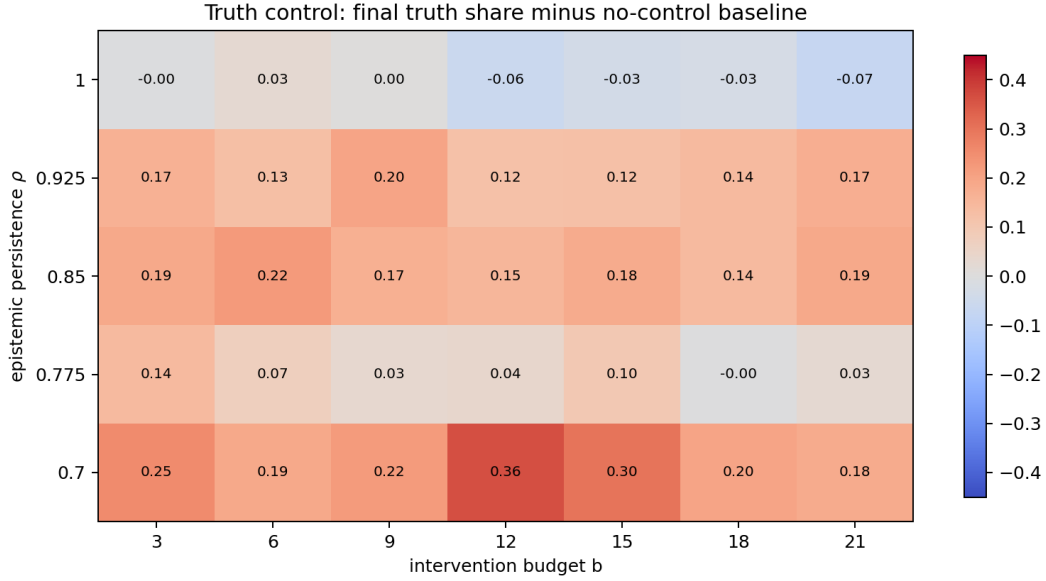


Figure 3: Truth-control final truth share minus the no-control baseline at the same persistence.

3.3 False control

False-directed control does not require false statements to alter the population. Aggregated over persistence, the false target is strongest at intermediate budgets. The episode-level final plurality rates are 34.7%, 41.7%, 56.3%, 29.8%, 36.2%, 52.1%, and 44.9% for $b = 3, 6, 9, 12, 15, 18, 21$ respectively.

At $\rho = 0.70, b = 9$, the false target reaches a mean final share of 0.574 and is the final plurality in 7/9 available episodes. Another strong cell is $\rho = 0.925, b = 18$, where the mean false-target

share is 0.532 and it is the final plurality in 6/9 episodes. These localized pockets are substantially stronger than the global average and establish a clearly non-monotone b response.

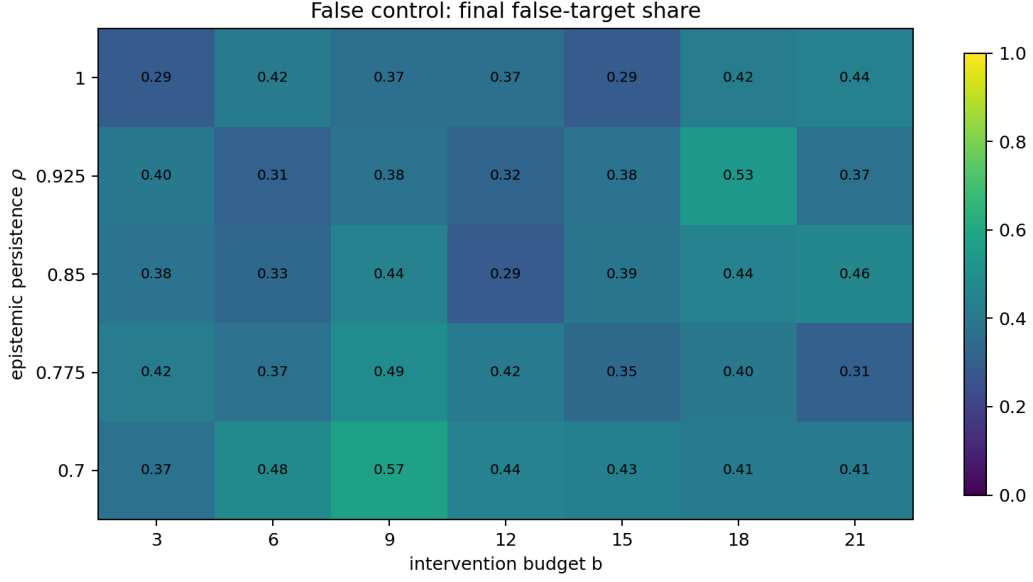


Figure 4: Final configured false-target share.

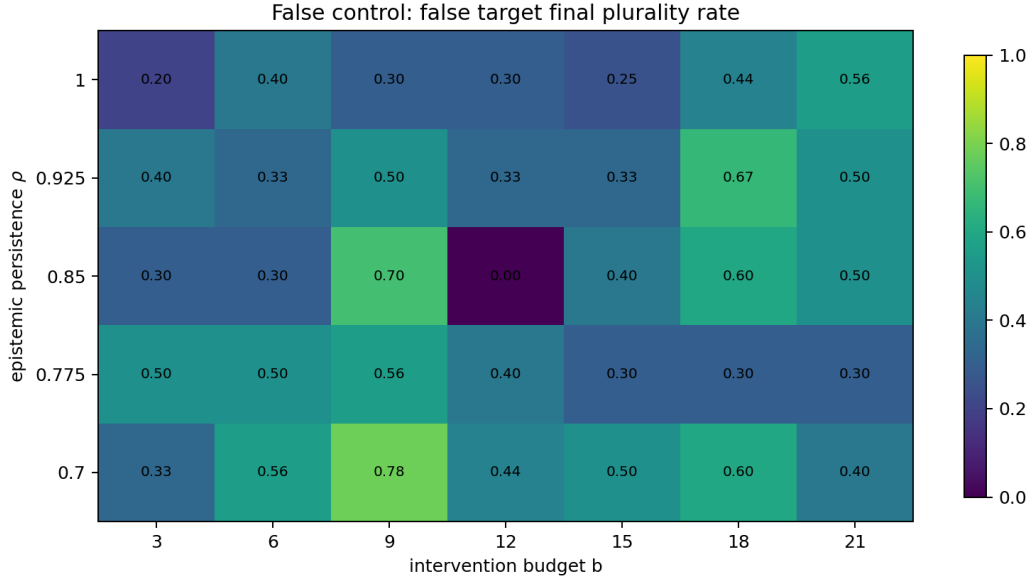


Figure 5: Fraction of episodes in which the false target is the final plurality.

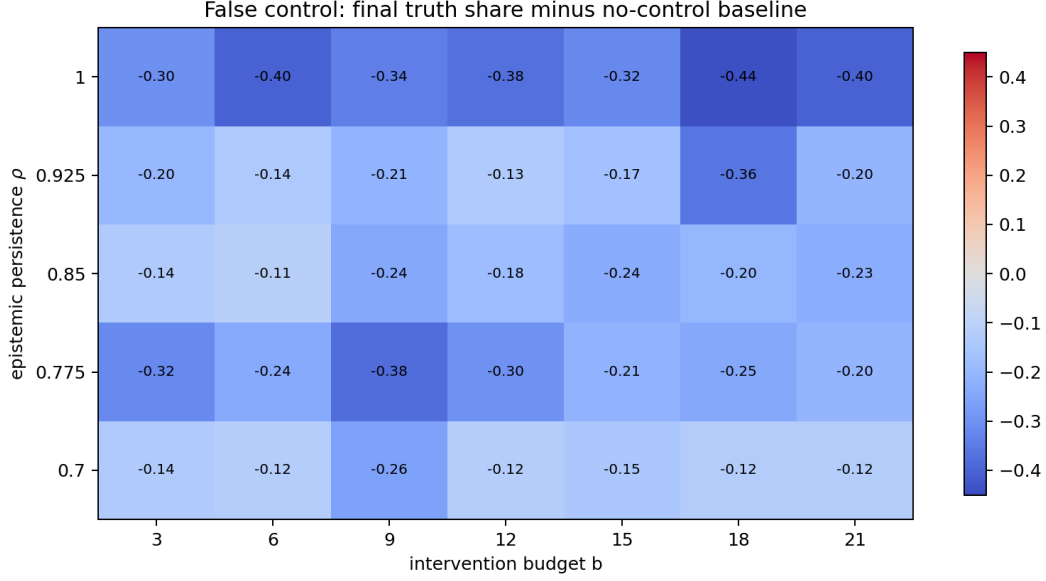


Figure 6: False-control final truth share minus the no-control truth baseline. Negative values indicate suppression of truth.

3.4 Pooled trajectories

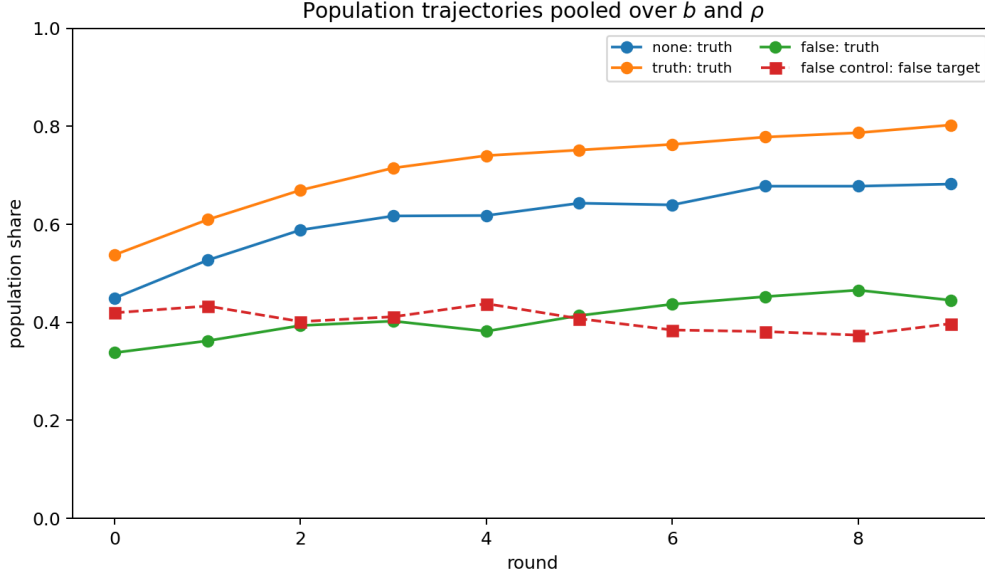


Figure 7: Round trajectories pooled over budgets and persistence. Truth control rapidly raises truth. False control holds the false target near 0.4 while substantially slowing the autonomous growth of truth.

The binary policy acts much more often in the false arm because the false target remains below the policy threshold more often. Across all controlled rounds, $P(U = 1) \approx 0.594$ under false control and 0.389 under truth control. This is an endogenous property of the same binary feedback policy, not a

configuration mismatch, but it means raw truth-vs-false comparisons combine target semantics with different realized intervention frequencies.

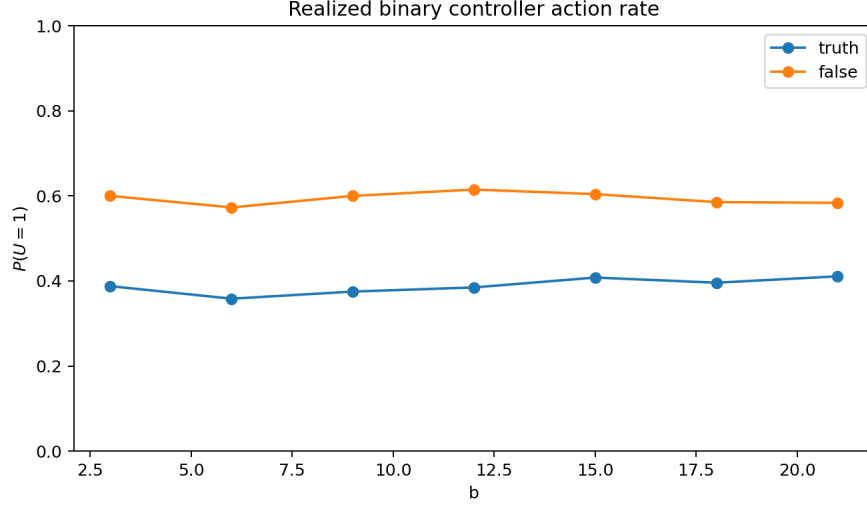


Figure 8: Realized binary controller action frequency versus budget.

4 Information and response metrics

4.1 Raw transfer information T_π and its permutation null

Raw target CMI is large across all cells. Mean raw T_π is 0.447 bits for truth control and 0.563 bits for false control.

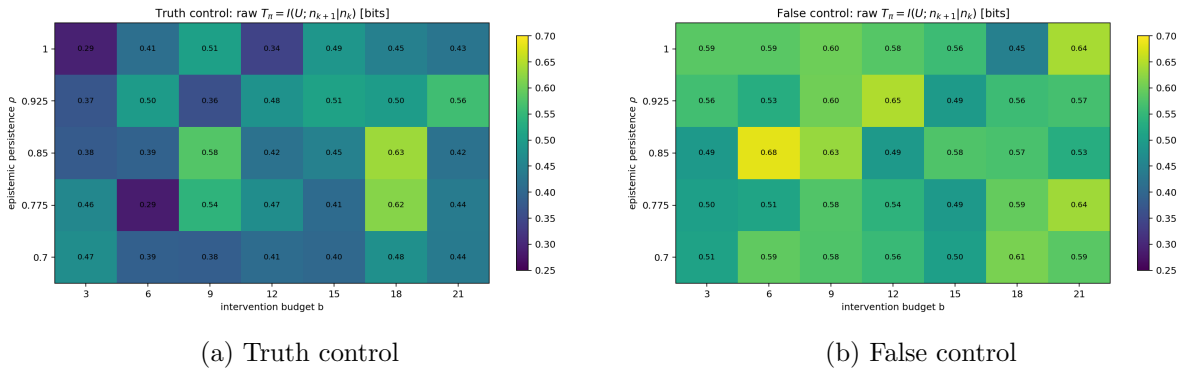


Figure 9: Whole-cell raw T_π .

However, the direct-count estimator is operating with roughly 15–23 occupied conditioning states from only 80–100 rounds per cell. The permutation null is therefore strongly positive. Mean null CMI is 0.470 bits for truth control and 0.527 bits for false control. The mean excess over the null is -0.0228 bits for truth control and +0.0366 bits for false control.

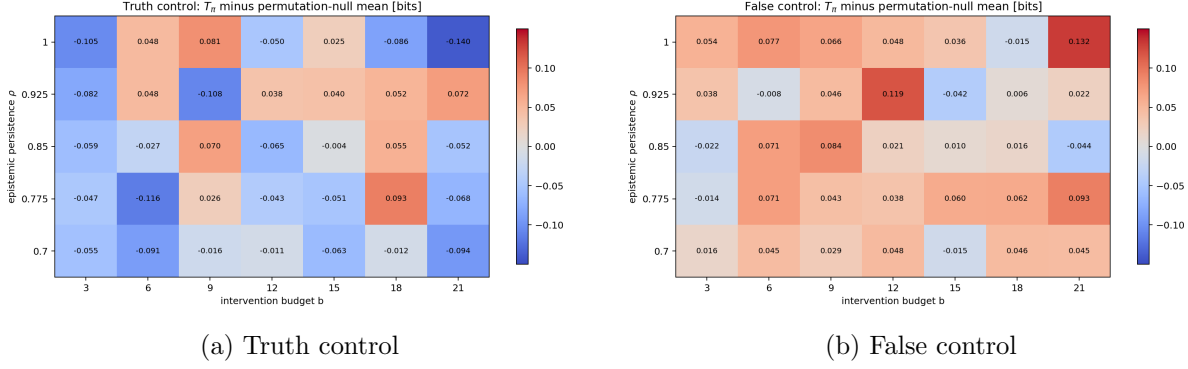


Figure 10: Raw T_π minus the permutation-null mean. This is a diagnostic, not a redefinition of T_π .

Only two false-control cells have permutation $p < 0.05$: approximately $(\rho, b) = (1.0, 21)$ and $(0.925, 12)$. No truth-control cell passes $p < 0.05$. Consequently, the positive bootstrap interval of the raw nonnegative CMI estimator should not be interpreted as evidence against independence. The permutation test is the relevant inferential check here.

4.2 η_{IF}

The reported information fraction is large: mean $\eta_{IF} = 0.678$ for truth control and 0.778 for false control. Since this quantity uses the same raw actuation information in its numerator, its absolute magnitude inherits the finite-sample CMI concern above. It remains useful as a descriptive surface, but should not yet be treated as a calibrated fraction of genuine causal information transfer.

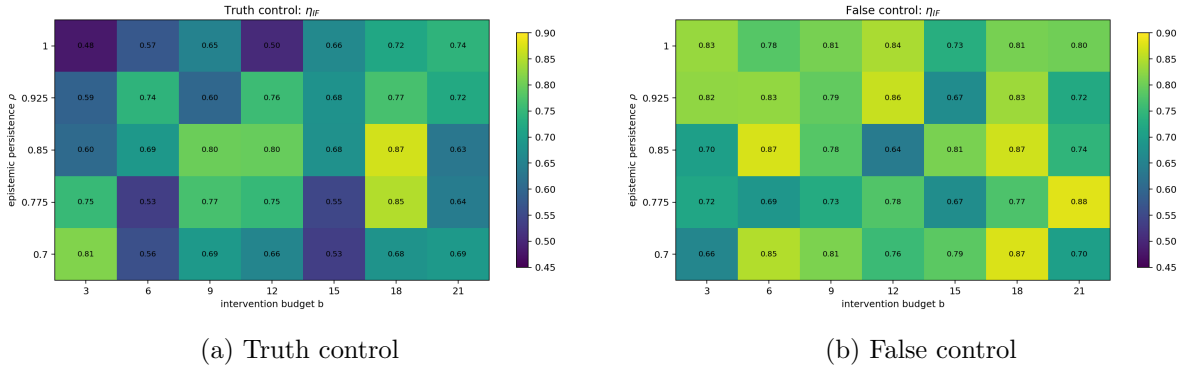


Figure 11: Whole-cell η_{IF} .

4.3 Susceptibility χ

Susceptibility provides the clearest same-round response signal in this dataset. The mean target-fraction response is 0.0272 under truth control and 0.0875 under false control. A positive 95% bootstrap interval occurs in 26/35 false cells and 4/35 truth cells.

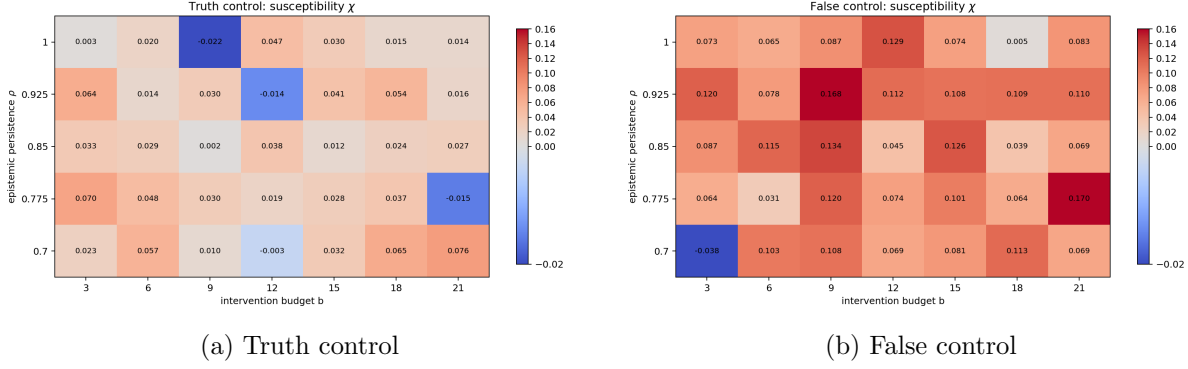


Figure 12: Whole-cell susceptibility χ .

The stronger false-arm susceptibility is plausible: the false target usually has more available population mass to gain, whereas truth is often already being amplified by autonomous reasoning and peer evidence. It should therefore not be read as “false evidence is intrinsically more powerful”; it is a state- and policy-dependent response.

4.4 η_{IR}

Occupancy-weighted η_{IR} averages 0.0105 for truth control and 0.0193 for false control. Since the denominator is raw T_π , an upward-biased CMI denominator tends to depress η_{IR} . These values are therefore best treated as provisional until the actuation-information estimator is stabilized.

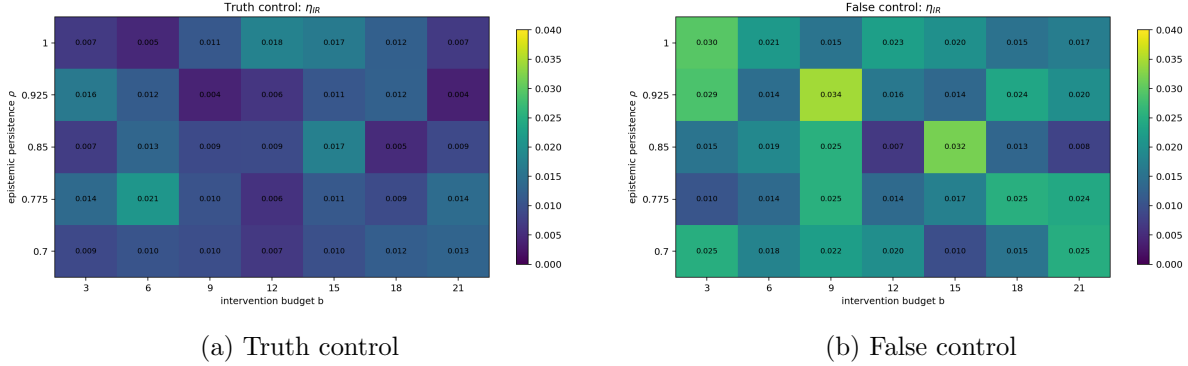


Figure 13: Whole-cell η_{IR} .

5 State-resolved $x \times b$ structure

The supplied archive’s pre-rendered state-local phase diagrams are blank because every row in `state_local_phase_maps.parquet` is stamped `structural_cell_not_run`. This is an aggregation/plotting-layer problem rather than absence of data: `primary_estimates.parquet` contains 1,980 x -resolved primary-estimator rows. The maps below were reconstructed directly from those estimates and show only cells marked `adequate`; unsupported cells are left blank.

5.1 Truth control

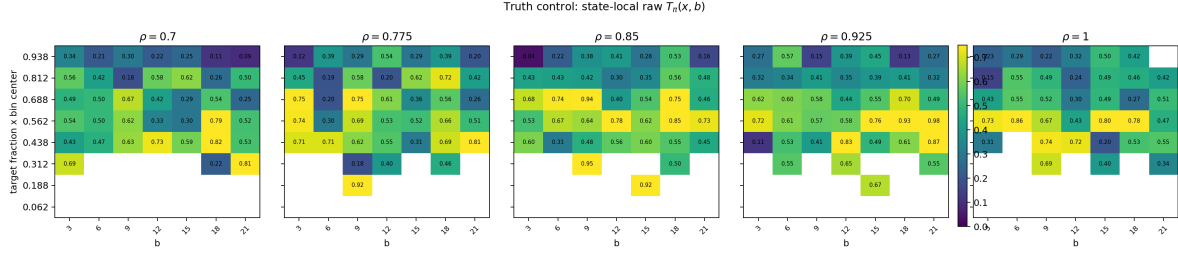


Figure 14: Truth control: raw state-local $T_\pi(x, b)$.

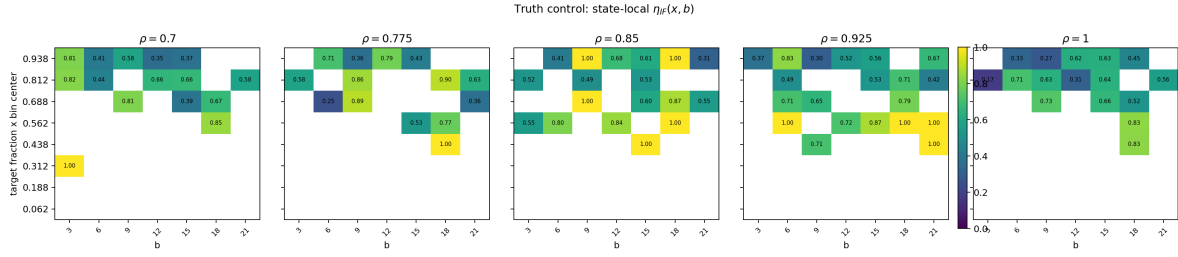


Figure 15: Truth control: state-local $\eta_{IF}(x, b)$.

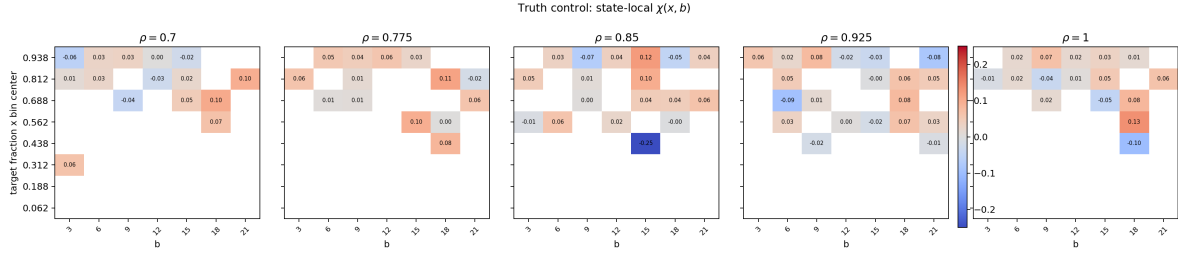


Figure 16: Truth control: state-local $\chi(x, b)$.

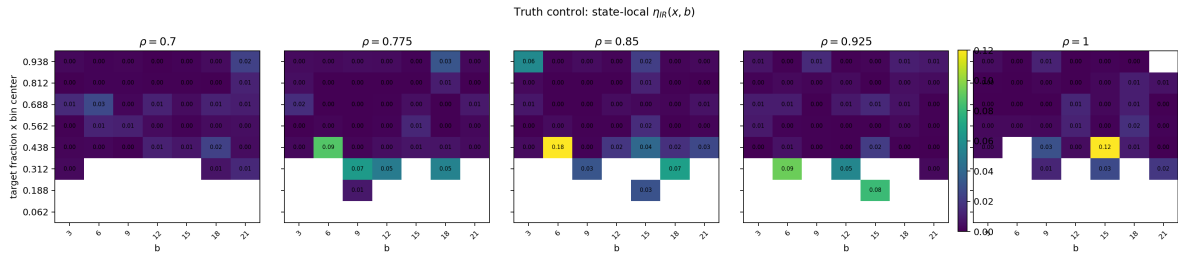


Figure 17: Truth control: state-local $\eta_{IR}(x, b)$.

5.2 False control

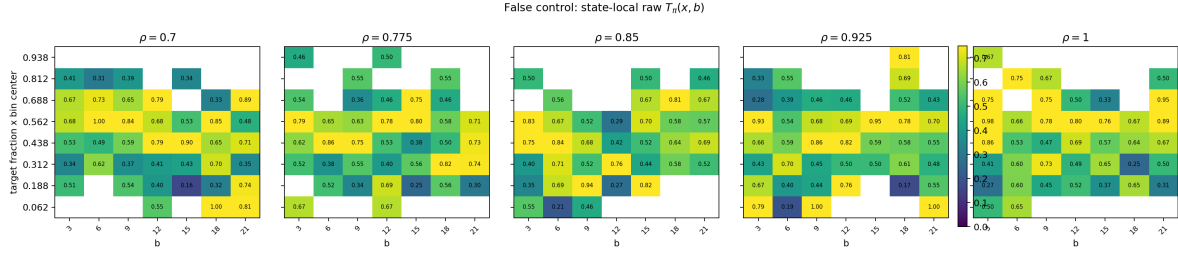


Figure 18: False control: raw state-local $T_\pi(x, b)$.

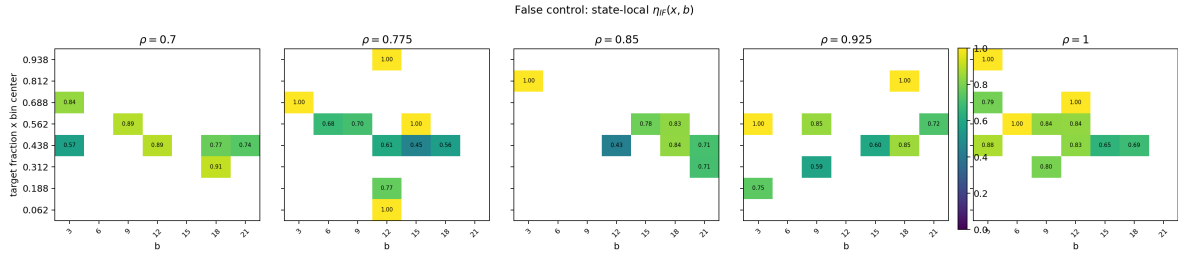


Figure 19: False control: state-local $\eta_{IF}(x, b)$.

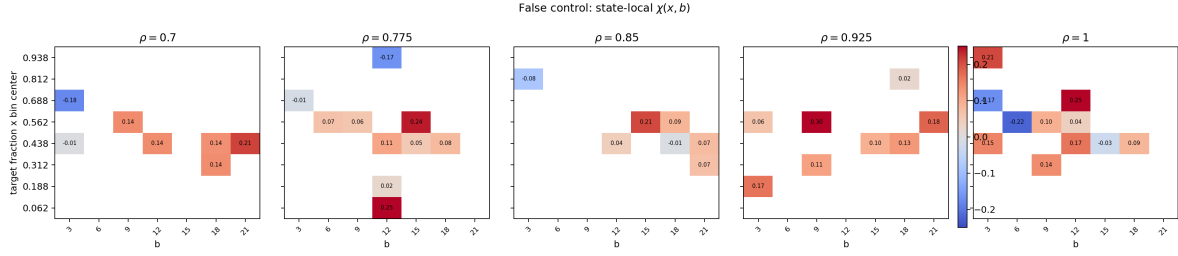


Figure 20: False control: state-local $\chi(x, b)$.

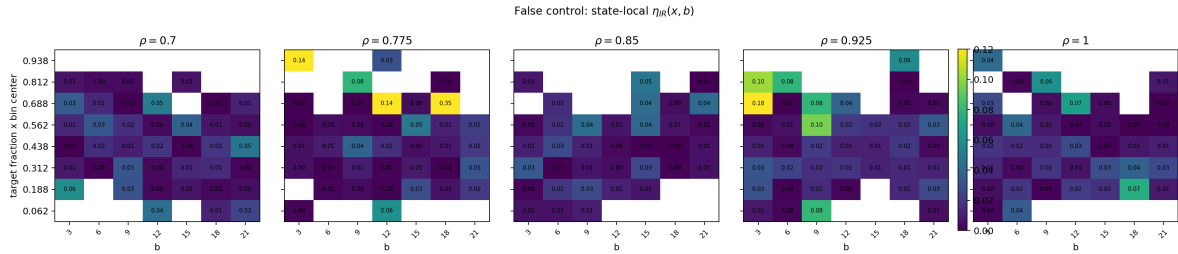


Figure 21: False control: state-local $\eta_{IR}(x, b)$.

The state-local surfaces are sparse because each cell contains only 8–10 episodes and 10 rounds. They nonetheless reinforce the conclusion that response is localized in (x, b, ρ) rather than monotone in

budget. In particular, false-control χ contains several positive response pockets around intermediate target fractions, together with occasional negative cells, which is consistent with a nonlinear controlled kernel rather than simple additive persuasion.

6 Blackboard mechanism and control efficiency

6.1 Communication funnel

On ACT rounds the controller always admits the requested number of reports. Increasing b strongly increases total exposure, but the downstream behavioral yield saturates.

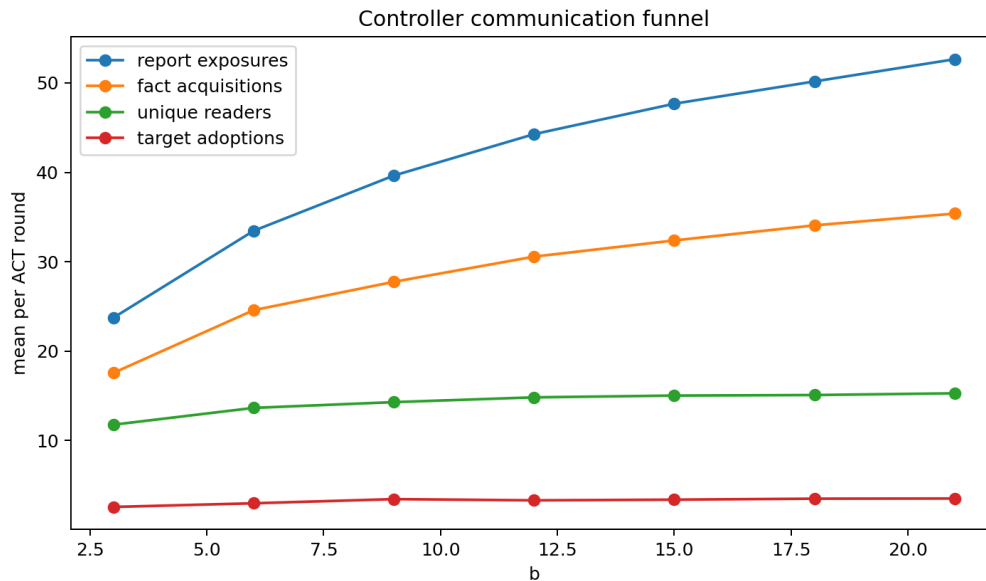


Figure 22: Mean controller communication funnel per ACT round, pooled across persistence and target semantics.

At $b = 3$, the controller obtains about 23.7 report exposures and 11.8 unique readers per ACT round; at $b = 21$ these become about 52.7 exposures and 15.3 unique readers. Thus the number of unique people reached saturates much earlier than the number of message reads.

6.2 Marginal yield

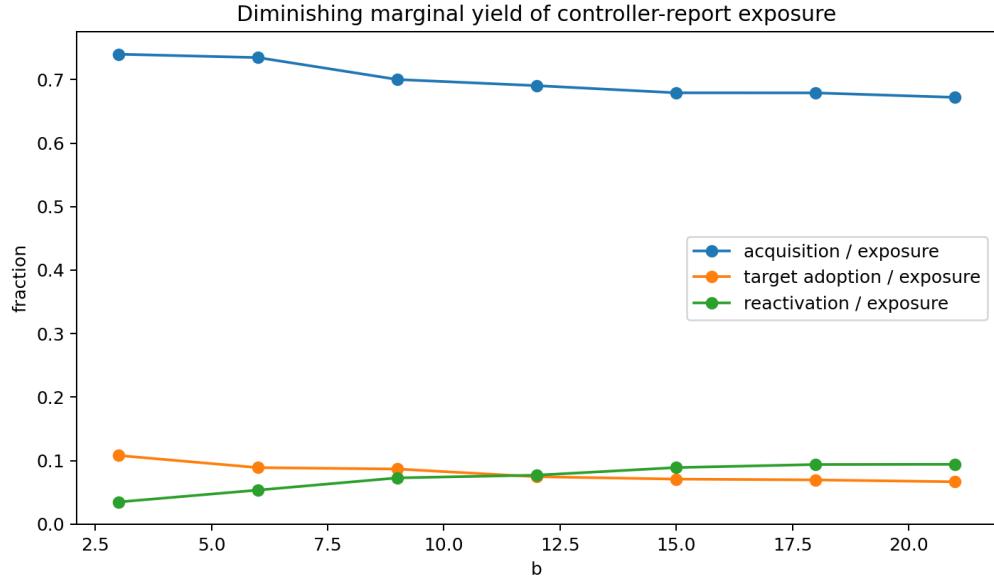


Figure 23: Per-exposure downstream yield.

The number of exposures per controller post falls from about 7.9 at $b = 3$ to 2.5 at $b = 21$. Exact fact acquisition per exposure falls from roughly 0.74 to 0.67, while target adoption per exposure falls from about 0.108 to 0.067. Reactivation per exposure rises with b , indicating that high budgets increasingly revisit evidence already seen historically rather than creating entirely new knowledge.

6.3 Coverage versus crowding

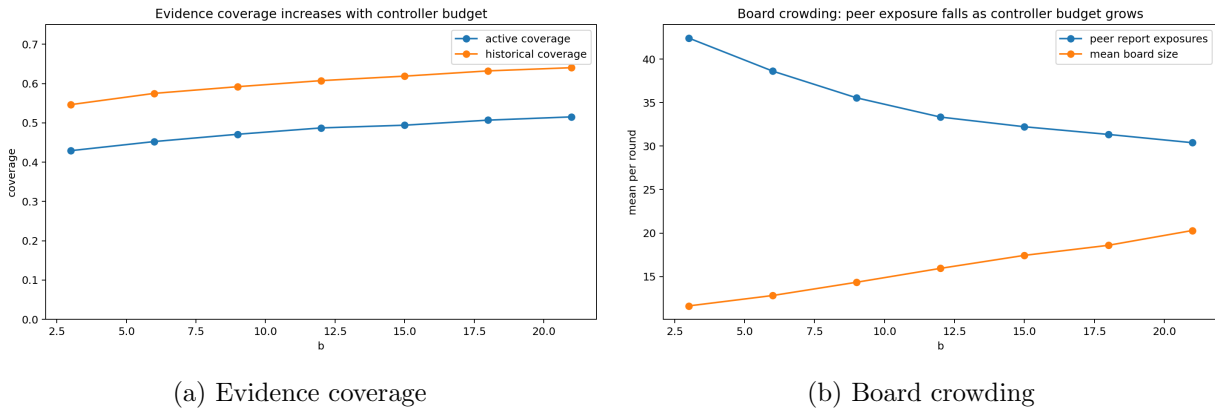


Figure 24: Higher control budget increases evidence coverage but reduces exposure to peer reports.

This is one of the clearest mechanistic results of the study. Controller budget is not simply “more control.” It trades off controller evidence against peer-to-peer information exchange. This likely explains part of the non-monotonic behavioral response with b and is directly relevant to control-efficiency interpretation.

7 Thermodynamic quantities

The archive correctly reports η_{th} as unsupported in all 75 cells. The reason is `insufficient_support_for_h_calibration`. The blackboard report actuator does not generate the direct microscopic $+/-$ transition statistics required by the old affinity calibration. Therefore:

$$\eta_{th}, \quad \eta_{th}^{\text{signed}}, \quad \eta_{th}^{\text{bounded}}$$

are all undefined in this study. This should not be filled heuristically. A blackboard-compatible effective-affinity derivation or calibration is required before thermodynamic efficiency can be reported.

8 Interpretation for the next experimental stage

This run is scientifically encouraging for the newer program. A controller restricted to truthful evidence can exert substantial control, including adversarial steering, and does so through a measurable communication funnel rather than repeated factless directives. At the same time, the results make clear that communication budget has two competing effects: it increases controller information exposure and memory coverage, but crowds out peer communication and produces diminishing adoption yield.

The most important next steps are:

1. Complete the 30 missing episodes before using this grid for final paper values.
2. Keep the raw T_π surface, but always accompany it with the permutation null or a null-corrected diagnostic. With 10 episodes per cell, exact-state direct-count CMI is strongly biased upward.
3. Consider either more repetitions, longer horizons, coarser state conditioning, or a bias-reduced conditional-information estimator before making quantitative claims from T_π or η_{IF} .
4. Preserve χ as a primary immediate-response observable; it is much more convincingly identified in the current sample.
5. Proceed with the planned cooperation experiment in which ordinary-agent REQUEST capability can be turned on/off. The very low REQUEST rate here shows that spontaneous questioning is not yet a strong component of the dynamics.
6. In the adaptive-controller study, keep U binary and treat REPORT/REQUEST/DIRECTIVE as internal realizations of the controlled kernel Q_1 , preserving the theoretical information bound.

9 Bottom line

The main behavioral result is not a simple monotonic phase transition in b or ρ . Instead, truthful report control produces a **structured, non-monotone controllability landscape**. False-target steering is strongest in localized budget/persistence regions, while high communication budgets show clear saturation and crowding. The most convincing same-round control statistic in the present sample is susceptibility χ . The raw information-transfer metrics are large but currently dominated by a large finite-sample permutation baseline, so they require more data or estimator refinement before being used as strong quantitative evidence.